CLASS 10 CHEMISTRY FOUNDATION

Complete Exam-Ready Notes

Structured with Quick Summaries · Real-Life Examples · Formula Sheets
Memory Hooks · Common Mistakes · Chapter-End Revision

Chapter 1: Chemical Reactions and Equations

Prerequisites

Before starting this chapter, make sure you are comfortable with the following ideas:

- Knowledge of chemical symbols, formulae and valency.
- Difference between a physical and a chemical change.
- Idea of the law of conservation of mass.
- Familiarity with the terms element, compound and mixture.

1.1 Chemical Equations and Balancing

⏱ **Estimated time:** 15 mins **Difficulty:** 🟡 Medium ★★☆ **Exam Importance:** ⭐⭐⭐⭐⭐

🔑 Quick Summary

A chemical equation is the symbolic representation of a reaction, and it must be balanced so that the number of atoms of each element is the same on both sides.

🌍 Real-Life Example

A rusting gate, a burning candle and digesting food are all reactions that a chemist can write down in one line each.

📖 Detailed Explanation

Definition

A chemical equation represents a chemical reaction using symbols and formulae, with reactants on the left and products on the right. A balanced equation has an equal number of atoms of each element on both sides.

Key Idea

Balancing is required by the law of conservation of mass: matter is neither created nor destroyed in a chemical reaction. This is why only coefficients may be changed, never the subscripts in a formula.

Working Principle

The hit-and-trial method balances the element occurring in the largest number of atoms first, leaving hydrogen and oxygen for last. Physical states — (s), (l), (g), (aq) — and conditions such as heat, catalyst or pressure are written afterwards to make the equation informative.

Applications

- $Fe + H_2O \rightarrow Fe_3O_4 + H_2$ balances to $3Fe + 4H_2O \rightarrow Fe_3O_4 + 4H_2$.
- The '↑' sign shows a gas evolved and '↓' a precipitate formed.
- Reaction conditions are written above or below the arrow, e.g. Δ for heat.

✓ Points to Remember (Exam Focus)

- ✓ Reactants on the left, products on the right, arrow shows direction.
- ✓ A balanced equation obeys the law of conservation of mass.
- ✓ Only coefficients may be changed while balancing, NEVER subscripts.
- ✓ Physical states (s), (l), (g), (aq) make an equation more informative.
- ✓ Balance the element with the most atoms first; balance H and O last.

⚠ Common Student Mistakes

- ⚠ Changing a subscript to balance an equation, which changes the substance itself.
- ⚠ Forgetting to check every element after adjusting one coefficient.
- ⚠ Omitting physical states when the question asks for a complete equation.

Memory Hook

'Coefficients yes, subscripts never' — and 'metal first, hydrogen and oxygen last'.

Advanced Insights

A balanced equation is also a statement about moles: $2H_2 + O_2 \rightarrow 2H_2O$ says two moles of hydrogen react with one of oxygen. This is what makes stoichiometric calculation possible from the equation alone.

1.2 Types of Chemical Reactions

⏱ **Estimated time:** 17 mins **Difficulty:** 🟡 Medium ★★☆ **Exam Importance:** ⭐⭐⭐⭐⭐

Quick Summary

Reactions are classified as combination, decomposition, displacement, double displacement and redox, and separately as exothermic or endothermic by the energy change.

Real-Life Example

Quicklime added to water gets hot enough to boil it, while a cold pack for a sprain gets icy — one exothermic, one endothermic.

📖 Detailed Explanation

Definition

Combination: two or more substances form one product. Decomposition: one compound splits into two or more. Displacement: a more reactive element displaces a less reactive one. Double displacement: two compounds exchange ions.

Key Idea

Decomposition is the exact reverse of combination and always requires energy — supplied as heat (thermal), electricity (electrolytic) or light (photolytic).

Working Principle

In a double displacement reaction that produces an insoluble solid, the reaction is called precipitation; if it occurs between an acid and a base, it is neutralisation.

Applications

- Combination: $CaO + H_2O \rightarrow Ca(OH)_2$, strongly exothermic (slaking of lime).
- Decomposition: $2AgBr \rightarrow 2Ag + Br_2$ in sunlight — used in black-and-white photography.
- Displacement: $Fe + CuSO_4 \rightarrow FeSO_4 + Cu$; blue solution turns green.

✓ Points to Remember (Exam Focus)

- ✓ Thermal decomposition: $CaCO_3 \rightarrow CaO + CO_2$ on heating.
- ✓ Electrolytic decomposition: $2H_2O \rightarrow 2H_2 + O_2$ using electricity.
- ✓ Photolytic decomposition: silver halides break down in sunlight.
- ✓ Precipitation: $Na_2SO_4 + BaCl_2 \rightarrow BaSO_4\downarrow + 2NaCl$ (white precipitate).
- ✓ Exothermic reactions release heat; endothermic reactions absorb it.

⚠ Common Student Mistakes

- ⚠ Confusing displacement (one element replaced) with double displacement (ions exchanged).
- ⚠ Assuming all decomposition reactions need heat; some need light or electricity.
- ⚠ Calling respiration endothermic; it is exothermic.

Memory Hook

'Combine, Decompose, Displace, Double-displace' — and 'decomposition always demands energy'.

Advanced Insights

Respiration is exothermic yet occurs at body temperature because enzymes lower the activation energy. Energy release and reaction rate are separate questions — thermodynamics says whether, kinetics says how fast.

Quick Recap — Topics Covered So Far

- Chemical Equations and Balancing — A chemical equation is the symbolic representation of a reaction, and it must be balanced so that the number of atoms of each element is the same on both sides.
- Types of Chemical Reactions — Reactions are classified as combination, decomposition, displacement, double displacement and redox, and separately as exothermic or endothermic by the energy change.

1.3 Oxidation, Reduction and Redox Reactions

⏱ **Estimated time:** 14 mins **Difficulty:** 🟡 Medium ★★☆ **Exam Importance:** ⭐⭐⭐⭐⭐

🔎 **Quick Summary**

Oxidation is gain of oxygen or loss of hydrogen and reduction is the reverse; because they always occur together the combined process is called a redox reaction.

🌍 **Real-Life Example**

A shiny copper vessel turning dull black over a flame is copper being oxidised — and rubbing it with hydrogen would reduce it right back.

📖 Detailed Explanation

Definition

Oxidation is the addition of oxygen or removal of hydrogen (or loss of electrons). Reduction is the addition of hydrogen or removal of oxygen (or gain of electrons). A reaction in which both occur simultaneously is a redox reaction.

Key Idea

The substance that supplies oxygen is the oxidising agent and is itself reduced; the substance that removes oxygen is the reducing agent and is itself oxidised. The agent always undergoes the opposite of what it causes.

Working Principle

In $CuO + H_2 \rightarrow Cu + H_2O$, copper oxide loses oxygen and is reduced, while hydrogen gains oxygen and is oxidised. CuO is therefore the oxidising agent and H_2 the reducing agent.

Applications

- Corrosion: iron rusts, silver tarnishes black, copper develops a green coating.
- Rancidity: oxidation of fats and oils gives food a bad smell and taste.
- Rancidity is prevented by antioxidants, nitrogen flushing and refrigeration.

✓ Points to Remember (Exam Focus)

✓ Oxidation = gain of oxygen / loss of hydrogen / loss of electrons.

✓ Reduction = loss of oxygen / gain of hydrogen / gain of electrons.

- ✓ The oxidising agent is itself REDUCED; the reducing agent is itself OXIDISED.
- ✓ Corrosion is prevented by painting, oiling, galvanising, anodising and alloying.
- ✓ Chips packets are flushed with nitrogen to prevent oxidative rancidity.

⚠ Common Student Mistakes

- ⚠ Saying the oxidising agent gets oxidised; it gets reduced.
- ⚠ Treating oxidation and reduction as separate reactions; they always occur together.
- ⚠ Thinking oxidation always needs oxygen; loss of electrons counts even without it.

Memory Hook

'OIL RIG' — Oxidation Is Loss of electrons, Reduction Is Gain — and 'the agent does the opposite to itself'.

Advanced Insights

Galvanising protects iron by sacrificial corrosion: zinc is more readily oxidised, so it corrodes preferentially even at a scratch. Tin plating, being less reactive than iron, actually accelerates rusting once the coating breaks.

Quick Recap — Topics Covered So Far

- Oxidation, Reduction and Redox Reactions — Oxidation is gain of oxygen or loss of hydrogen and reduction is the reverse; because they always occur together the combined process is called a redox reaction.

Chapter 1 — End of Chapter Revision

All Memory Hooks

- Chemical Equations and Balancing: 'Coefficients yes, subscripts never' — and 'metal first, hydrogen and oxygen last'.
- Types of Chemical Reactions: 'Combine, Decompose, Displace, Double-displace' — and 'decomposition always demands energy'.
- Oxidation, Reduction and Redox Reactions: 'OIL RIG' — Oxidation Is Loss of electrons, Reduction Is Gain — and 'the agent does the opposite to itself'.

⚠ All Common Mistakes

⚠ Changing a subscript to balance an equation, which changes the substance itself.

⚠ Forgetting to check every element after adjusting one coefficient.

⚠ Omitting physical states when the question asks for a complete equation.

⚠ Confusing displacement (one element replaced) with double displacement (ions exchanged).

⚠ Assuming all decomposition reactions need heat; some need light or electricity.

⚠ Calling respiration endothermic; it is exothermic.

⚠ Saying the oxidising agent gets oxidised; it gets reduced.

⚠ Treating oxidation and reduction as separate reactions; they always occur together.

⚠ Thinking oxidation always needs oxygen; loss of electrons counts even without it.

Quick Revision Section

- Chemical Equations and Balancing: A chemical equation is the symbolic representation of a reaction, and it must be balanced so that the number of atoms of each element is the same on both sides.
- Types of Chemical Reactions: Reactions are classified as combination, decomposition, displacement, double displacement and redox, and separately as exothermic or endothermic by the energy change.
- Oxidation, Reduction and Redox Reactions: Oxidation is gain of oxygen or loss of hydrogen and reduction is the reverse; because they always occur together the combined process is called a redox reaction.

Chapter 2: Acids, Bases and Salts

Prerequisites

Before starting this chapter, make sure you are comfortable with the following ideas:

- Idea of ions and radicals from the language of chemistry.
- Familiarity with litmus paper and indicators.
- Knowledge of common acids such as HCl and H_2SO_4.
- Understanding of a neutralisation reaction.

2.1 Properties of Acids and Bases

⏱ **Estimated time:** 15 mins **Difficulty:** 🟢 Easy ★☆☆ **Exam Importance:** ⭐⭐⭐⭐⭐

🔑 Quick Summary

Acids produce H^+ ions in water and turn blue litmus red, bases produce OH^- ions and turn red litmus blue, and indicators are used to tell them apart.

🌍 Real-Life Example

Lemon juice turns blue litmus red because of citric acid, while soap solution turns red litmus blue because it is basic.

📖 Detailed Explanation

Definition

An acid is a substance that produces hydrogen ions (H^+) in aqueous solution; a base produces hydroxide ions (OH^-). A base soluble in water is called an alkali. An indicator is a substance that shows a distinct colour change in acid and base.

Key Idea

Acids show acidic behaviour only in aqueous solution, because they need water to release H^+ ions. Dry HCl gas does not turn dry blue litmus red — a classic examination demonstration.

Working Principle

Acid + metal → salt + hydrogen; acid + metal carbonate → salt + CO_2 + water; acid + base → salt + water (neutralisation); metal oxide + acid → salt + water.

Applications

- Natural indicators: litmus, turmeric, red cabbage, china rose petals.
- Olfactory indicators such as onion and vanilla essence change smell rather than colour.
- Hydrogen gas is tested by bringing a burning candle near — it burns with a pop sound.

✓ Points to Remember (Exam Focus)

- ✓ Acids: sour, turn blue litmus red, release H^+ in water, react with metals to give H_2.
- ✓ Bases: bitter, soapy to touch, turn red litmus blue, release OH^- in water.
- ✓ Acids show acidic behaviour ONLY in aqueous solution.
- ✓ Phenolphthalein: colourless in acid, pink in base. Methyl orange: red in acid, yellow in base.
- ✓ CO_2 turns lime water milky — the standard test for carbon dioxide.

⚠ Common Student Mistakes

- ⚠ Saying dry HCl gas turns dry litmus red; water is essential.
- ⚠ Adding water to concentrated acid instead of acid to water, which can cause violent splashing.
- ⚠ Assuming all bases are alkalis; only water-soluble bases are alkalis.

Memory Hook

'Acid Adds H^+, Base Brings OH^-' — and 'Always Add Acid to water, never the reverse'.

Advanced Insights

Dilution of a concentrated acid is strongly exothermic because hydration of the ions releases a large amount of energy. Adding water to acid concentrates that heat in a small volume, which can boil the water and splash acid out.

2.2 The pH Scale and Its Importance

⏱ **Estimated time:** 14 mins **Difficulty:** 🟡 Medium ★★☆ **Exam Importance:** ⭐⭐⭐⭐⭐

Quick Summary

The pH scale from 0 to 14 measures the concentration of hydrogen ions, with 7 neutral, below 7 acidic and above 7 basic.

Real-Life Example

Toothpaste is basic because it neutralises the acid produced by bacteria in the mouth, which attacks enamel below pH 5.5.

📖 Detailed Explanation

Definition

pH is a number that indicates the acidic or basic nature of a solution; it is inversely related to the concentration of hydrogen ions. The 'p' in pH stands for 'potenz', meaning power in German.

Key Idea

The scale is logarithmic, so each unit represents a tenfold change. A solution of pH 3 is not slightly more acidic than pH 5 but a hundred times more acidic.

Working Principle

A strong acid ionises completely and produces many H^+ ions, giving a very low pH. A weak acid ionises only partly, so its solution has a higher pH at the same concentration.

Applications

- Human blood is maintained between pH 7.35 and 7.45; a small deviation is dangerous.
- Plants grow best in soil close to neutral pH; acidic soil is treated with quicklime.
- Bee sting is acidic and relieved by baking soda; wasp sting is basic and relieved by vinegar.

✓ Points to Remember (Exam Focus)

- ✓ pH < 7 acidic, pH = 7 neutral, pH > 7 basic.
- ✓ The scale is logarithmic: one unit means a tenfold change in H^+ concentration.
- ✓ Tooth decay begins when the mouth pH falls below 5.5.
- ✓ Stomach acid is HCl with pH about 1.2–3; antacids such as milk of magnesia neutralise it.
- ✓ Acid rain has pH below 5.6 and damages aquatic life and buildings.

⚠ Common Student Mistakes

- ⚠ Treating the pH scale as linear — the difference between pH 3 and pH 5 is a factor of 100.
- ⚠ Confusing strength (degree of ionisation) with concentration (amount dissolved).
- ⚠ Saying pH 0 or pH 14 is impossible; both are attainable in strong solutions.

Memory Hook

'Below 7 sour, above 7 soapy, 7 exactly neutral' — and each step is a factor of ten.

Advanced Insights

Blood pH is held within a tenth of a unit by the bicarbonate buffer system, which absorbs added acid or base. Buffering, not avoidance, is how biology copes with constant chemical disturbance.

⏸ Quick Recap — Topics Covered So Far

- Properties of Acids and Bases — Acids produce H^+ ions in water and turn blue litmus red, bases produce OH^- ions and turn red litmus blue, and indicators are used to tell them apart.
- The pH Scale and Its Importance — The pH scale from 0 to 14 measures the concentration of hydrogen ions, with 7 neutral, below 7 acidic and above 7 basic.

2.3 Salts and Their Preparation

⏱ **Estimated time:** 16 mins **Difficulty:** 🟡 Medium ★★☆ **Exam Importance:** ⭐⭐⭐⭐⭐

Quick Summary

Salts are formed by neutralisation of an acid with a base, and their pH depends on whether the parent acid and base were strong or weak.

Real-Life Example

Common salt is the raw material for baking soda, washing soda, bleaching powder and caustic soda — an entire industry from one mineral.

Detailed Explanation

Definition

A salt is an ionic compound formed by the reaction of an acid with a base, in which the hydrogen of the acid is replaced by a metal or an ammonium ion.

Key Idea

Strong acid + strong base gives a neutral salt ($NaCl$, pH 7). Strong acid + weak base gives an acidic salt (NH_4Cl). Weak acid + strong base gives a basic salt (Na_2CO_3).

Working Principle

The chlor-alkali process electrolyses brine to give sodium hydroxide at the cathode, chlorine at the anode and hydrogen as a by-product. All three products have major industrial uses.

Applications

- Baking soda $NaHCO_3$: antacid, baking powder ingredient, soda-acid fire extinguisher.
- Washing soda $Na_2CO_3 \cdot 10H_2O$: removes permanent hardness of water, used in glass and paper.
- Bleaching powder $CaOCl_2$: disinfecting drinking water, bleaching cotton and wood pulp.

✓ Points to Remember (Exam Focus)

- ✓ Chlor-alkali process: brine → $NaOH$ (cathode) + Cl_2 (anode) + H_2.
- ✓ Plaster of Paris: $CaSO_4 \cdot \frac{1}{2}H_2O$, made by heating gypsum to 373 K; sets on adding water.
- ✓ Baking soda is a mild, non-corrosive base that decomposes on heating to give CO_2.
- ✓ Water of crystallisation is the fixed number of water molecules in a crystal, e.g. $CuSO_4 \cdot 5H_2O$.

✓ Blue copper sulphate crystals turn white on heating and blue again on adding water.

⚠ Common Student Mistakes

- ⚠ Writing plaster of Paris as $CaSO_4 \cdot 2H_2O$; that is gypsum, not plaster of Paris.
- ⚠ Assuming all salts are neutral; the parent acid and base decide the pH.
- ⚠ Confusing baking soda ($NaHCO_3$) with washing soda ($Na_2CO_3 \cdot 10H_2O$).

Memory Hook

'Strong beats weak' — whichever parent is stronger decides whether the salt is acidic or basic.

Advanced Insights

Plaster of Paris must be stored in a moisture-proof container because it sets by rehydrating back to gypsum. The setting is a chemical reaction, not drying, which is why it hardens even under water.

⏸ Quick Recap — Topics Covered So Far

- Salts and Their Preparation — Salts are formed by neutralisation of an acid with a base, and their pH depends on whether the parent acid and base were strong or weak.

Chapter 2 — End of Chapter Revision

All Memory Hooks

- Properties of Acids and Bases: 'Acid Adds H^+, Base Brings OH^-' — and 'Always Add Acid to water, never the reverse'.
- The pH Scale and Its Importance: 'Below 7 sour, above 7 soapy, 7 exactly neutral' — and each step is a factor of ten.
- Salts and Their Preparation: 'Strong beats weak' — whichever parent is stronger decides whether the salt is acidic or basic.

⚠ All Common Mistakes

⚠ Saying dry HCl gas turns dry litmus red; water is essential.

⚠ Adding water to concentrated acid instead of acid to water, which can cause violent splashing.

⚠ Assuming all bases are alkalis; only water-soluble bases are alkalis.

⚠ Treating the pH scale as linear — the difference between pH 3 and pH 5 is a factor of 100.

⚠ Confusing strength (degree of ionisation) with concentration (amount dissolved).

⚠ Saying pH 0 or pH 14 is impossible; both are attainable in strong solutions.

⚠ Writing plaster of Paris as $CaSO_4 \cdot 2H_2O$; that is gypsum, not plaster of Paris.

⚠ Assuming all salts are neutral; the parent acid and base decide the pH.

⚠ Confusing baking soda ($NaHCO_3$) with washing soda ($Na_2CO_3 \cdot 10H_2O$).

Quick Revision Section

- Properties of Acids and Bases: Acids produce H^+ ions in water and turn blue litmus red, bases produce OH^- ions and turn red litmus blue, and indicators are used to tell them apart.
- The pH Scale and Its Importance: The pH scale from 0 to 14 measures the concentration of hydrogen ions, with 7 neutral, below 7 acidic and above 7 basic.
- Salts and Their Preparation: Salts are formed by neutralisation of an acid with a base, and their pH depends on whether the parent acid and base were strong or weak.

Chapter 3: Metals and Non-Metals

Prerequisites

Before starting this chapter, make sure you are comfortable with the following ideas:

- Physical properties of metals and non-metals from earlier classes.
- Idea of valency and electronic configuration.
- Understanding of oxidation and reduction.
- Familiarity with the reactivity series.

3.1 Physical and Chemical Properties

⏱ **Estimated time:** 15 mins **Difficulty:** 🟢 Easy ★☆☆ **Exam Importance:** ⭐⭐⭐⭐⭐

Quick Summary

Metals are lustrous, malleable, ductile and good conductors and form basic oxides, while non-metals are brittle, poor conductors and form acidic oxides.

Real-Life Example

Aluminium foil can be beaten into a sheet thinner than paper, while a stick of sulphur crumbles at a tap — the same hammer, opposite results.

Detailed Explanation

Definition

Malleability is the ability to be beaten into sheets; ductility is the ability to be drawn into wires; sonorousness is the ability to produce a ringing sound. Amphoteric oxides react with both acids and bases.

Key Idea

Metals lose electrons to form positive ions and their oxides are basic; non-metals gain electrons to form negative ions and their oxides are acidic. Every property in this chapter follows from that one difference in electron behaviour.

Working Principle

Metals react with oxygen to form oxides, with water to form hydroxide and hydrogen, and with dilute acids to release hydrogen — but only metals above hydrogen in the reactivity series do the last.

Applications

- Aluminium and zinc form amphoteric oxides, reacting with both acids and alkalis.
- Aqua regia (3 : 1 concentrated HCl : HNO_3) is the only reagent that dissolves gold and platinum.
- Anodising thickens the natural oxide layer on aluminium to protect it further.

✓ Points to Remember (Exam Focus)

- ✓ Exceptions: mercury is liquid, sodium and potassium are soft, graphite conducts, iodine is lustrous, diamond is hardest.
- ✓ Metal oxides are basic; non-metal oxides are acidic; Al_2O_3 and ZnO are amphoteric.
- ✓ Metals below hydrogen in the reactivity series (Cu, Ag, Au) do not release H_2 from dilute acids.
- ✓ Nitric acid does not usually give hydrogen with metals — it oxidises it to water.
- ✓ Aqua regia dissolves gold and platinum; it is a 3:1 mixture of conc. HCl and conc. HNO_3.

⚠ Common Student Mistakes

- ⚠ Saying all metals react with dilute acids to give hydrogen; copper, silver and gold do not.
- ⚠ Expecting hydrogen gas with nitric acid; it is normally reduced to nitrogen oxides instead.
- ⚠ Calling all metal oxides basic and forgetting the amphoteric ones.

Memory Hook

'Metals give electrons and make bases; non-metals take electrons and make acids.'

Advanced Insights

Aluminium appears unreactive despite being high in the reactivity series because a thin, tightly adherent oxide layer forms instantly and seals the surface. Reactivity and corrosion resistance are therefore not the same property.

3.2 Ionic Compounds and the Reactivity Series

⏱ **Estimated time:** 16 mins **Difficulty:** 🟡 Medium ★★☆ **Exam Importance:** ⭐⭐⭐⭐⭐

Quick Summary

Metals transfer electrons to non-metals to form ionic compounds, which are hard, high-melting and conduct electricity when molten or dissolved.

Real-Life Example

Solid table salt does not conduct electricity, but the moment it dissolves in water the solution completes a circuit — the ions have been set free.

📖 Detailed Explanation

Definition

An ionic or electrovalent bond is formed by the complete transfer of one or more electrons from a metal atom to a non-metal atom. The reactivity series lists metals in decreasing order of reactivity.

Key Idea

Ionic compounds do not conduct in the solid state because the ions are locked in a rigid lattice. Melting or dissolving frees the ions, and only then can charge move.

Working Principle

In a displacement reaction, a metal higher in the reactivity series displaces one lower down from its salt solution. The same ordering decides which method must be used to extract each metal from its ore.

Applications

- Sodium (2,8,1) loses one electron and chlorine (2,8,7) gains one, forming Na^+Cl^-.
- Iron displaces copper from copper sulphate solution; copper cannot displace iron.
- Thermite reaction: $Fe_2O_3 + 2Al \rightarrow 2Fe + Al_2O_3$, used for welding railway tracks.

✓ Points to Remember (Exam Focus)

- ✓ Reactivity series: K > Na > Ca > Mg > Al > Zn > Fe > Pb > H > Cu > Hg > Ag > Au.
- ✓ Ionic compounds: high melting and boiling points, hard, brittle, soluble in water.
- ✓ They conduct electricity in molten or aqueous state, NOT in the solid state.
- ✓ A more reactive metal displaces a less reactive metal from its salt solution.
- ✓ The thermite reaction is highly exothermic and produces molten iron.

⚠ Common Student Mistakes

- ⚠ Saying ionic compounds conduct in the solid state; the ions cannot move.
- ⚠ Reversing displacement — the more reactive metal displaces the less reactive one.
- ⚠ Assuming all ionic compounds dissolve in water; some, like $BaSO_4$, do not.

Memory Hook

'Please Send Cats, Monkeys And Zebras In Lead Cages Made Securely Gold-plated' — the reactivity series in order.

Advanced Insights

The reactivity series is an empirical ordering of standard electrode potentials, which measure how readily a metal gives up electrons. The same table therefore predicts displacement, extraction method and galvanic corrosion.

Quick Recap — Topics Covered So Far

- Physical and Chemical Properties — Metals are lustrous, malleable, ductile and good conductors and form basic oxides, while non-metals are brittle, poor conductors and form acidic oxides.
- Ionic Compounds and the Reactivity Series — Metals transfer electrons to non-metals to form ionic compounds, which are hard, high-melting and conduct electricity when molten or dissolved.

3.3 Extraction of Metals and Corrosion

⏱ **Estimated time:** 16 mins **Difficulty:** 🟡 Medium ★★☆ **Exam Importance:** ⭐⭐⭐⭐⭐

Quick Summary

The method of extracting a metal depends on its position in the reactivity series, and corrosion is prevented by coating, alloying or sacrificial protection.

Real-Life Example

Iron railings need repainting every few years, but the Iron Pillar at Delhi has resisted rust for over sixteen centuries because of its unusual composition.

Detailed Explanation

Definition

Metallurgy is the process of extracting a metal from its ore and refining it. Roasting is heating a sulphide ore strongly in excess air; calcination is heating a carbonate ore in limited air.

Key Idea

The extraction method follows the reactivity series exactly. Highly reactive metals must be extracted by electrolysis, moderately reactive ones by reduction with carbon, and least reactive ones simply by heating their ore in air.

Working Principle

Concentration removes gangue, roasting or calcination converts the ore to oxide, reduction gives the free metal, and electrolytic refining yields the pure metal at the cathode.

Applications

- Aluminium is obtained from bauxite by electrolysis (Hall–Héroult process).
- Zinc and iron oxides are reduced by carbon or carbon monoxide in a furnace.
- Mercury and copper can be obtained simply by heating their sulphide ores in air.

✓ Points to Remember (Exam Focus)

✓ Roasting: sulphide ore + excess air → oxide + SO_2. Calcination: carbonate ore, limited air → oxide + CO_2.

✓ Highly reactive metals (K, Na, Ca, Mg, Al): extracted by electrolysis of the molten ore.

✓ Moderately reactive metals (Zn, Fe, Pb): oxide reduced by carbon.

✓ Least reactive metals (Cu, Ag, Hg): obtained by heating the ore in air alone.

✓ Rusting requires BOTH air and moisture; galvanising, painting and alloying prevent it.

⚠ Common Student Mistakes

⚠ Confusing roasting (sulphides, excess air) with calcination (carbonates, limited air).

⚠ Trying to reduce aluminium oxide with carbon; aluminium is too reactive for that.

⚠ Saying rusting occurs in dry air; both oxygen and water are needed.

Memory Hook

'Reactive → Electrolysis, Middling → Carbon, Lazy → just Heat it' — extraction follows the series.

Advanced Insights

Alloying changes properties fundamentally: adding chromium and nickel to iron gives stainless steel, whose chromium oxide layer self-repairs when scratched. That self-healing film, not hardness, is what makes it stainless.

Quick Recap — Topics Covered So Far

- Extraction of Metals and Corrosion — The method of extracting a metal depends on its position in the reactivity series, and corrosion is prevented by coating, alloying or sacrificial protection.

Chapter 3 — End of Chapter Revision

🧠 All Memory Hooks

- Physical and Chemical Properties: 'Metals give electrons and make bases; non-metals take electrons and make acids.'
- Ionic Compounds and the Reactivity Series: 'Please Send Cats, Monkeys And Zebras In Lead Cages Made Securely Gold-plated' — the reactivity series in order.
- Extraction of Metals and Corrosion: 'Reactive → Electrolysis, Middling → Carbon, Lazy → just Heat it' — extraction follows the series.

⚠ All Common Mistakes

⚠ Saying all metals react with dilute acids to give hydrogen; copper, silver and gold do not.

⚠ Expecting hydrogen gas with nitric acid; it is normally reduced to nitrogen oxides instead.

⚠ Calling all metal oxides basic and forgetting the amphoteric ones.

⚠ Saying ionic compounds conduct in the solid state; the ions cannot move.

⚠ Reversing displacement — the more reactive metal displaces the less reactive one.

⚠ Assuming all ionic compounds dissolve in water; some, like $BaSO_4$, do not.

⚠ Confusing roasting (sulphides, excess air) with calcination (carbonates, limited air).

⚠ Trying to reduce aluminium oxide with carbon; aluminium is too reactive for that.

⚠ Saying rusting occurs in dry air; both oxygen and water are needed.

⚡ Quick Revision Section

- Physical and Chemical Properties: Metals are lustrous, malleable, ductile and good conductors and form basic oxides, while non-metals are brittle, poor conductors and form acidic oxides.
- Ionic Compounds and the Reactivity Series: Metals transfer electrons to non-metals to form ionic compounds, which are hard, high-melting and conduct electricity when molten or dissolved.
- Extraction of Metals and Corrosion: The method of extracting a metal depends on its position in the reactivity series, and corrosion is prevented by coating, alloying or sacrificial protection.

Chapter 4: Carbon and Its Compounds

Prerequisites

Before starting this chapter, make sure you are comfortable with the following ideas:

- Idea of electronic configuration and valency.
- Difference between ionic and covalent bonding.
- Knowledge that carbon has atomic number 6.
- Familiarity with fuels such as LPG and petrol.

4.1 Covalent Bonding, Catenation and Allotropes

⏱ **Estimated time:** 16 mins **Difficulty:** 🟡 Medium ★★☆ **Exam Importance:** ⭐⭐⭐⭐⭐

🔑 Quick Summary

Carbon shares electrons to form four covalent bonds, and its ability to bond with itself — catenation — together with tetravalency explains the millions of carbon compounds.

🌍 Real-Life Example

Diamond and graphite are both pure carbon, yet one cuts glass and the other writes on paper — the difference is entirely in how the atoms are joined.

📖 Detailed Explanation

Definition

A covalent bond is formed by the mutual sharing of electrons between two atoms. Catenation is the ability of an element to form long chains, branches and rings by bonding with its own atoms.

Key Idea

Carbon cannot lose four electrons (too much energy needed) or gain four (the nucleus could not hold ten electrons), so it must share. That single constraint produces the whole of organic chemistry.

Working Principle

Carbon–carbon bonds are exceptionally strong because carbon is small and the shared pair is held close to both nuclei. This strength makes long chains stable, which silicon — the next element down — cannot manage.

Applications

- Diamond: each carbon bonded to four others in a rigid 3D lattice — hardest natural substance.
- Graphite: layers of hexagonal rings with one free electron per atom — soft and conducting.
- Fullerene (C_{60}, buckminsterfullerene) has a football-like cage structure.

✓ Points to Remember (Exam Focus)

- ✓ Carbon is tetravalent: it forms four covalent bonds.
- ✓ Covalent compounds: low melting and boiling points, poor conductors, generally insoluble in water.
- ✓ Catenation plus tetravalency explains the enormous number of carbon compounds.
- ✓ Diamond is the hardest substance; graphite is a good conductor and a lubricant.
- ✓ Allotropes are different physical forms of the same element in the same physical state.

⚠ Common Student Mistakes

- ⚠ Saying covalent compounds conduct electricity; they generally do not, as they have no free ions.
- ⚠ Calling diamond and graphite different compounds; they are allotropes of the same element.
- ⚠ Thinking carbon forms C^{4+} or C^{4-} ions in its compounds.

Memory Hook

'Four bonds, endless chains' — tetravalency plus catenation is the whole story.

Advanced Insights

Graphite conducts because each atom uses only three of its four valence electrons in bonding, leaving one delocalised within the layer. The same free electrons make graphene an outstanding conductor.

4.2 Hydrocarbons and Homologous Series

⏱ **Estimated time:** 16 mins **Difficulty:** 🟡 Medium ★★☆ **Exam Importance:** ⭐⭐⭐⭐⭐

Quick Summary

Hydrocarbons are compounds of carbon and hydrogen, saturated if all bonds are single and unsaturated if they contain a double or triple bond, and they form homologous series.

Real-Life Example

The methane in cooking gas, the ethene that ripens fruit and the acetylene in a welding torch are the first three members of three different families.

📖 Detailed Explanation

Definition

Saturated hydrocarbons (alkanes) contain only single bonds; unsaturated hydrocarbons contain a double bond (alkenes) or a triple bond (alkynes). A homologous series is a family of compounds with the same general formula in which consecutive members differ by a $-CH_2-$ unit.

Key Idea

All members of a homologous series have the same functional group and similar chemical properties, while physical properties change gradually with molecular mass. This is why learning one member effectively teaches the whole family.

Working Principle

Saturated hydrocarbons burn with a clean blue flame; unsaturated ones burn with a yellow sooty flame because their higher carbon percentage means incomplete combustion.

Applications

- Alkanes C_nH_{2n+2}: methane, ethane, propane, butane — used as fuels.
- Alkenes C_nH_{2n}: ethene ripens fruit; alkynes C_nH_{2n-2}: ethyne used in welding.
- Functional groups: −OH alcohol, −CHO aldehyde, >C=O ketone, −COOH carboxylic acid.

✓ Points to Remember (Exam Focus)

- ✓ Alkane C_nH_{2n+2} (single bonds), alkene C_nH_{2n} (double bond), alkyne C_nH_{2n-2} (triple bond).
- ✓ Consecutive members of a homologous series differ by CH_2 and by 14 u in molecular mass.
- ✓ Isomers have the same molecular formula but different structures; butane onwards show isomerism.
- ✓ Saturated hydrocarbons burn with a clean blue flame; unsaturated ones with a sooty yellow flame.
- ✓ IUPAC names use a prefix for chain length, a suffix for the functional group.

⚠ Common Student Mistakes

- ⚠ Writing the alkyne general formula as C_nH_{2n-1} instead of C_nH_{2n-2}.
- ⚠ Saying methane and ethane are isomers; isomers must have the SAME molecular formula.
- ⚠ Claiming the first three alkanes show isomerism; isomerism starts at butane.

Memory Hook

'Meth, Eth, Prop, But, Pent, Hex' — and '-ane single, -ene double, -yne triple'.

Advanced Insights

The homologous series exists because chemistry is governed by the functional group while physical properties are governed by molecular size. This separation is why an entire family can be predicted from a single member.

II Quick Recap — Topics Covered So Far

- Covalent Bonding, Catenation and Allotropes — Carbon shares electrons to form four covalent bonds, and its ability to bond with itself — catenation — together with tetravalency explains the millions of carbon compounds.
- Hydrocarbons and Homologous Series — Hydrocarbons are compounds of carbon and hydrogen, saturated if all bonds are single and unsaturated if they contain a double or triple bond, and they form homologous series.

4.3 Chemical Properties of Carbon Compounds

⏱ **Estimated time:** 15 mins **Difficulty:** 🟡 Medium ★★☆ **Exam Importance:** ⭐⭐⭐⭐⭐

🔎 Quick Summary

Carbon compounds undergo combustion, oxidation, addition and substitution reactions, each characteristic of a particular type of bond.

🌍 Real-Life Example

Vegetable oil is hardened into vanaspati ghee by adding hydrogen across its double bonds — an addition reaction on an industrial scale.

📖 Detailed Explanation

Definition

An addition reaction is one in which atoms are added across a double or triple bond. A substitution reaction is one in which an atom or group is replaced by another. Oxidation converts alcohols to carboxylic acids.

Key Idea

Unsaturated compounds undergo addition because they have a bond to spare; saturated compounds can only undergo substitution, and even that requires sunlight. Bond type dictates reaction type.

Working Principle

Hydrogenation of vegetable oil uses nickel as a catalyst and converts unsaturated oils into saturated fats. Substitution of methane by chlorine requires sunlight and gives a mixture of chlorinated products.

Applications

- Hydrogenation: oil + H_2 (Ni catalyst) → vanaspati ghee.
- Ethanol + alkaline $KMnO_4$ or acidified $K_2Cr_2O_7$ → ethanoic acid (oxidation).
- Esterification: ethanol + ethanoic acid (conc. H_2SO_4) → ester with a sweet fruity smell.

✓ Points to Remember (Exam Focus)

✓ Addition reactions occur in unsaturated compounds; substitution in saturated ones.

✓ Nickel and palladium act as catalysts in hydrogenation.

✓ Alkaline $KMnO_4$ and acidified $K_2Cr_2O_7$ are oxidising agents that convert alcohol to acid.

✓ Esterification gives a sweet-smelling ester; saponification is the reverse with alkali, giving soap.

✓ Animal fats are saturated and vegetable oils unsaturated; unsaturated oils are healthier.

⚠ Common Student Mistakes

- ⚠ Expecting an addition reaction from an alkane; alkanes have no spare bond.
- ⚠ Forgetting that substitution of methane needs sunlight.
- ⚠ Confusing esterification (acid + alcohol → ester) with saponification (ester + alkali → soap).

Memory Hook

'Unsaturated Adds, Saturated Substitutes' — the bond decides the reaction.

Advanced Insights

Partial hydrogenation produces trans fats, whose straight chains pack tightly and raise the risk of heart disease. The same reaction that improved shelf life created a public-health problem, and many countries now restrict it.

4.4 Ethanol, Ethanoic Acid, Soaps and Detergents

⏱ **Estimated time:** 16 mins **Difficulty:** 🟡 Medium ★★☆ **Exam Importance:** ⭐⭐⭐⭐⭐

Quick Summary

Ethanol and ethanoic acid are the two most important carbon compounds in the syllabus, and soaps clean by having one end that dissolves in water and another that dissolves in grease.

Real-Life Example

Soap fails to lather in hard water and forms a grey scum instead, which is exactly why detergents were invented.

📖 Detailed Explanation

Definition

Soap is the sodium or potassium salt of a long-chain carboxylic acid. A micelle is a spherical cluster in which the hydrophobic tails point inward towards the oil and the hydrophilic heads point outward towards the water.

Key Idea

A soap molecule has two ends with opposite affinities — an ionic head that dissolves in water and a hydrocarbon tail that dissolves in grease. Cleaning is simply this dual nature at work.

Working Principle

In hard water, calcium and magnesium ions react with soap to form an insoluble scum, wasting soap. Detergents use sulphonate or sulphate heads whose calcium and magnesium salts are soluble, so they lather even in hard water.

Applications

- Ethanol is used in alcoholic drinks, as a solvent, in tinctures and as a fuel additive.
- Ethanoic acid (5–8% solution) is vinegar; glacial acetic acid freezes in cold climates.
- Detergents are used in shampoos and washing powders because they work in hard water.

✓ Points to Remember (Exam Focus)

✓ Ethanol C_2H_5OH: reacts with sodium to give hydrogen; dehydrated by conc. H_2SO_4 at 443 K to give ethene.

✓ Ethanoic acid CH_3COOH: melting point 290 K, so it freezes in cold climates — hence 'glacial'.

✓ Ethanoic acid + sodium carbonate → sodium ethanoate + CO_2 + water (brisk effervescence).

✓ Soap: sodium salt of a fatty acid; ineffective in hard water because it forms scum.

✓ Detergents work in hard water because their calcium and magnesium salts are soluble.

⚠ Common Student Mistakes

⚠ Saying detergents are biodegradable like soaps; many detergents are not.

⚠ Confusing denatured alcohol (made unfit to drink) with glacial acetic acid.

⚠ Thinking soap cleans by dissolving grease in water; it forms micelles that suspend the grease.

Memory Hook

'Head loves water, Tail loves grease' — that split personality is the whole of detergency.

Advanced Insights

Micelles keep grease suspended as an emulsion rather than dissolved, which is why rinsing carries the dirt away. Branched-chain detergents resist bacterial breakdown, which is why modern formulations use straight chains.

Quick Recap — Topics Covered So Far

- Chemical Properties of Carbon Compounds — Carbon compounds undergo combustion, oxidation, addition and substitution reactions, each characteristic of a particular type of bond.

- Ethanol, Ethanoic Acid, Soaps and Detergents — Ethanol and ethanoic acid are the two most important carbon compounds in the syllabus, and soaps clean by having one end that dissolves in water and another that dissolves in grease.

Chapter 4 — End of Chapter Revision

All Memory Hooks

- Covalent Bonding, Catenation and Allotropes: 'Four bonds, endless chains' — tetravalency plus catenation is the whole story.
- Hydrocarbons and Homologous Series: 'Meth, Eth, Prop, But, Pent, Hex' — and '-ane single, -ene double, -yne triple'.
- Chemical Properties of Carbon Compounds: 'Unsaturated Adds, Saturated Substitutes' — the bond decides the reaction.
- Ethanol, Ethanoic Acid, Soaps and Detergents: 'Head loves water, Tail loves grease' — that split personality is the whole of detergency.

⚠ All Common Mistakes

⚠ Saying covalent compounds conduct electricity; they generally do not, as they have no free ions.

⚠ Calling diamond and graphite different compounds; they are allotropes of the same element.

⚠ Thinking carbon forms C^{4+} or C^{4-} ions in its compounds.

⚠ Writing the alkyne general formula as C_nH_{2n-1} instead of C_nH_{2n-2}.

⚠ Saying methane and ethane are isomers; isomers must have the SAME molecular formula.

⚠ Claiming the first three alkanes show isomerism; isomerism starts at butane.

⚠ Expecting an addition reaction from an alkane; alkanes have no spare bond.

⚠ Forgetting that substitution of methane needs sunlight.

⚠ Confusing esterification (acid + alcohol → ester) with saponification (ester + alkali → soap).

⚠ Saying detergents are biodegradable like soaps; many detergents are not.

⚠ Confusing denatured alcohol (made unfit to drink) with glacial acetic acid.

⚠ Thinking soap cleans by dissolving grease in water; it forms micelles that suspend the grease.

Quick Revision Section

- Covalent Bonding, Catenation and Allotropes: Carbon shares electrons to form four covalent bonds, and its ability to bond with itself — catenation — together with tetravalency explains the millions of carbon compounds.
- Hydrocarbons and Homologous Series: Hydrocarbons are compounds of carbon and hydrogen, saturated if all bonds are single and unsaturated if they contain a double or triple bond, and they form homologous series.
- Chemical Properties of Carbon Compounds: Carbon compounds undergo combustion, oxidation, addition and substitution reactions, each characteristic of a particular type of bond.

- Ethanol, Ethanoic Acid, Soaps and Detergents: Ethanol and ethanoic acid are the two most important carbon compounds in the syllabus, and soaps clean by having one end that dissolves in water and another that dissolves in grease.

Chapter 5: Periodic Classification of Elements

Prerequisites

Before starting this chapter, make sure you are comfortable with the following ideas:

- Knowledge of atomic number, mass number and electronic configuration.
- Idea of valency and how it is determined.
- Difference between metals and non-metals.
- Familiarity with the symbols of common elements.

5.1 Early Attempts at Classification

⏱ **Estimated time:** 14 mins **Difficulty:** 🟡 Medium ★★☆ **Exam Importance:** ⭐⭐⭐⭐

🔎 Quick Summary

Döbereiner's triads and Newlands' law of octaves were the first attempts to order the elements, but each worked only for a small part of the known elements.

🌍 Real-Life Example

Ordering elements is like arranging a library — every early scheme worked for one shelf and failed on the next until the right principle was found.

📖 Detailed Explanation

Definition

Döbereiner's triads are groups of three elements with similar properties in which the atomic mass of the middle element is approximately the average of the other two. Newlands' law of octaves states that every eighth element has properties similar to the first.

Key Idea

Both schemes failed because they used atomic mass, which does not increase in a strictly regular way. The failures were as informative as the successes — they pointed towards a different organising property.

Working Principle

Döbereiner could identify only three triads among the elements known then. Newlands' law worked up to calcium and then broke down, and he had to place two elements in the same slot.

Applications

- Li, Na, K form a Döbereiner triad: $(7 + 39)/2 = 23$, the atomic mass of sodium.
- Newlands' law failed for elements beyond calcium and did not allow for undiscovered elements.
- These failures led directly to Mendeleev's use of properties rather than mass alone.

✓ Points to Remember (Exam Focus)

- ✓ Döbereiner's triads: only three could be identified among known elements.
- ✓ Newlands' law of octaves applied only up to calcium.
- ✓ Newlands assumed only 56 elements existed and no more would be discovered.
- ✓ Both classifications were based on atomic mass.
- ✓ The failure of these schemes showed that atomic mass is not the fundamental property.

⚠ Common Student Mistakes

- ⚠ Saying the law of octaves worked for all elements; it broke down after calcium.
- ⚠ Confusing Döbereiner (triads) with Newlands (octaves).
- ⚠ Thinking these were useless; each identified a genuine periodicity.

🧠 Memory Hook

'Döbereiner counted in Threes, Newlands counted in Eights' — both counted by mass, and both ran out.

🚀 Advanced Insights

Newlands drew his analogy from the musical octave, which invited ridicule at the time. The underlying periodicity he spotted was real; only the organising variable was wrong.

5.2 Mendeleev's Periodic Table

⏱ **Estimated time:** 15 mins **Difficulty:** 🟡 Medium ★★☆ **Exam Importance:** ⭐⭐⭐⭐⭐

🔑 Quick Summary

Mendeleev arranged elements by increasing atomic mass and similarity of properties, leaving gaps for undiscovered elements whose properties he predicted correctly.

🌍 Real-Life Example

Mendeleev predicted the properties of a then-unknown element he called eka-silicon; when germanium was discovered fifteen years later it matched almost exactly.

📖 Detailed Explanation

Definition

Mendeleev's periodic law states that the properties of elements are a periodic function of their atomic masses. His table had horizontal rows called periods and vertical columns called groups.

Key Idea

Mendeleev's genius was to give priority to properties over strict mass order, and to leave gaps rather than force elements into wrong places. Both decisions were vindicated by later discoveries.

Working Principle

He grouped elements by the formulae of their oxides and hydrides. Where the mass order conflicted with properties, he placed the element by its properties — as with cobalt and nickel.

Applications

- Gaps were left for eka-boron (scandium), eka-aluminium (gallium) and eka-silicon (germanium).
- Noble gases were discovered later and were accommodated in a new group without disturbing the table.
- Cobalt (58.9) was placed before nickel (58.7) on the basis of properties.

✓ Points to Remember (Exam Focus)

✓ Mendeleev's periodic law is based on ATOMIC MASS.

✓ He left gaps for undiscovered elements and predicted their properties.

✓ Noble gases were later added as a separate group without disturbing the arrangement.

✓ Limitation: no fixed position for hydrogen, which resembles both alkali metals and halogens.

✓ Limitation: isotopes have different masses but must occupy the same position.

⚠ Common Student Mistakes

⚠ Saying Mendeleev arranged elements by atomic number; that was Moseley's later correction.

⚠ Treating the gaps as errors; they were deliberate and predictive.

⚠ Forgetting that anomalous pairs such as Co/Ni were placed by properties, not mass.

Memory Hook

'Mass ordered it, Properties corrected it, Gaps proved it' — the three moves of Mendeleev.

Advanced Insights

The isotope problem was fatal to a mass-based table: chlorine-35 and chlorine-37 are chemically identical but differ in mass. Only when Moseley showed that atomic number is the fundamental property did the anomalies disappear.

Quick Recap — Topics Covered So Far

- Early Attempts at Classification — Döbereiner's triads and Newlands' law of octaves were the first attempts to order the elements, but each worked only for a small part of the known elements.

- Mendeleev's Periodic Table — Mendeleev arranged elements by increasing atomic mass and similarity of properties, leaving gaps for undiscovered elements whose properties he predicted correctly.

5.3 The Modern Periodic Table and Periodic Trends

⏱ **Estimated time:** 17 mins **Difficulty:** 🟡 Medium ★★☆ **Exam Importance:** ⭐⭐⭐⭐⭐

Quick Summary

The modern table arranges elements by increasing atomic number in 18 groups and 7 periods, and properties change in regular trends across periods and down groups.

Real-Life Example

Sodium reacts violently with water while its neighbour magnesium barely reacts at all — one step across the table changes chemistry completely.

Detailed Explanation

Definition

The modern periodic law states that the properties of elements are a periodic function of their atomic numbers. A group is a vertical column; a period is a horizontal row.

Key Idea

Elements in a group have the same number of valence electrons, which is why they behave alike chemically. Elements in a period have the same number of shells, which is why size and reactivity change steadily across it.

Working Principle

Across a period, nuclear charge increases while the number of shells stays constant, so atoms get smaller, metallic character decreases and non-metallic character increases. Down a group, a new shell is added each time, so atoms get larger and metallic character increases.

Applications

- Group 1 alkali metals, Group 17 halogens, Group 18 noble gases.
- Valency first increases from 1 to 4 and then decreases to 0 across a period.
- Metals lie on the left, non-metals on the right, and metalloids along the dividing staircase.

✓ Points to Remember (Exam Focus)

- ✓ Modern periodic table: 18 groups and 7 periods, arranged by atomic number.
- ✓ Atomic size DECREASES across a period and INCREASES down a group.
- ✓ Metallic character decreases across a period and increases down a group.
- ✓ Elements in the same group have the same number of valence electrons and the same valency.

✓ The number of the period equals the number of shells in the atom.

⚠ Common Student Mistakes

⚠ Saying atomic size increases across a period; it decreases because of rising nuclear charge.

⚠ Confusing group number with period number when locating an element.

⚠ Assuming the most reactive metal and most reactive non-metal are in the same region; they are at opposite corners.

Memory Hook

'Across you shrink, Down you grow' — and 'same group, same valence electrons, same behaviour'.

Advanced Insights

Atomic size shrinks across a period despite added electrons because each new electron enters the same shell while nuclear charge rises. Shielding stays roughly constant, so the effective nuclear pull on the outer electrons increases steadily.

Quick Recap — Topics Covered So Far

- The Modern Periodic Table and Periodic Trends — The modern table arranges elements by increasing atomic number in 18 groups and 7 periods, and properties change in regular trends across periods and down groups.

Chapter 5 — End of Chapter Revision

🧠 All Memory Hooks

- Early Attempts at Classification: 'Döbereiner counted in Threes, Newlands counted in Eights' — both counted by mass, and both ran out.
- Mendeleev's Periodic Table: 'Mass ordered it, Properties corrected it, Gaps proved it' — the three moves of Mendeleev.
- The Modern Periodic Table and Periodic Trends: 'Across you shrink, Down you grow' — and 'same group, same valence electrons, same behaviour'.

⚠ All Common Mistakes

⚠ Saying the law of octaves worked for all elements; it broke down after calcium.

⚠ Confusing Döbereiner (triads) with Newlands (octaves).

⚠ Thinking these were useless; each identified a genuine periodicity.

⚠ Saying Mendeleev arranged elements by atomic number; that was Moseley's later correction.

⚠ Treating the gaps as errors; they were deliberate and predictive.

⚠ Forgetting that anomalous pairs such as Co/Ni were placed by properties, not mass.

⚠ Saying atomic size increases across a period; it decreases because of rising nuclear charge.

⚠ Confusing group number with period number when locating an element.

⚠ Assuming the most reactive metal and most reactive non-metal are in the same region; they are at opposite corners.

⚡ Quick Revision Section

- Early Attempts at Classification: Döbereiner's triads and Newlands' law of octaves were the first attempts to order the elements, but each worked only for a small part of the known elements.
- Mendeleev's Periodic Table: Mendeleev arranged elements by increasing atomic mass and similarity of properties, leaving gaps for undiscovered elements whose properties he predicted correctly.
- The Modern Periodic Table and Periodic Trends: The modern table arranges elements by increasing atomic number in 18 groups and 7 periods, and properties change in regular trends across periods and down groups.

Chapter 6: Chemical Kinetics

Prerequisites

Before starting this chapter, make sure you are comfortable with the following ideas:

- Understanding of chemical reactions and balanced equations.
- Idea that particles are in constant motion.
- Familiarity with the term catalyst.
- Basic knowledge of concentration and temperature.

6.1 Rate of Reaction and Its Measurement

⏱ **Estimated time:** 15 mins **Difficulty:** 🟡 Medium ★★☆ **Exam Importance:** ⭐ ⭐ ⭐ ⭐ ⭐

🔑 Quick Summary

The rate of a reaction is the change in concentration of a reactant or product per unit time, and it decreases as the reaction proceeds and reactants are used up.

🌍 Real-Life Example

Milk sours in hours on a summer afternoon but takes days in a refrigerator — the same reaction running at two very different rates.

📖 Detailed Explanation

Definition

The rate of a chemical reaction is the change in concentration of a reactant or product per unit time. Average rate is measured over an interval; instantaneous rate is the rate at a particular moment.

Key Idea

Rate falls as a reaction proceeds because reactant concentration falls, so collisions become less frequent. This is why the initial rate is always the highest.

Working Principle

Rate is measured by following any property that changes measurably — volume of gas evolved, mass lost, colour intensity, pH or electrical conductivity — against time.

Applications

- Volume of CO_2 collected per minute when marble chips react with acid.
- Loss in mass of a flask as gas escapes during a reaction.
- Colour change followed with a colorimeter for reactions involving coloured species.

✓ Points to Remember (Exam Focus)

- ✓ Rate = change in concentration / time taken; unit mol L^{-1} s^{-1}.
- ✓ Rate is always taken as a positive quantity.
- ✓ The rate is highest at the start and decreases as reactants are consumed.
- ✓ For a reaction aA → bB, rate = −(1/a)d[A]/dt = +(1/b)d[B]/dt.
- ✓ Instantaneous rate is found from the slope of the tangent to a concentration–time graph.

⚠ Common Student Mistakes

- ⚠ Reporting a negative rate for a reactant; the negative sign is built into the definition.
- ⚠ Ignoring the stoichiometric coefficients when relating rates of different species.
- ⚠ Confusing average rate with instantaneous rate.

Memory Hook

'Rate = how fast the concentration changes' — highest at the start, falling ever after.

Advanced Insights

Reaction rates span an extraordinary range, from ionic reactions complete in nanoseconds to the rusting of iron over years. The same collision theory explains both — only the activation energy differs.

Formula Section

Average Rate of Reaction

Difficulty	🟢 Easy ★☆☆
Formula	Rate = Δ[concentration] / Δt
Meaning of Variables	Δ[concentration] = change in molar concentration, Δt = time interval in seconds.
When to Use	To find how fast a reactant is consumed or a product formed over a measured interval.
Common Mistakes	Forgetting to divide by the stoichiometric coefficient of the species chosen.
Shortcut / Memory Trick	Use the species easiest to measure, then convert using the balanced equation.

6.2 Factors Affecting the Rate of Reaction

⏱ **Estimated time:** 16 mins **Difficulty:** 🟡 Medium ★★☆ **Exam Importance:** ⭐⭐⭐⭐⭐

Quick Summary

Reaction rate depends on the nature of reactants, concentration, surface area, temperature, catalyst and, for gases, pressure.

Real-Life Example

Powdered sugar dissolves and burns far faster than a sugar cube, and a flour mill can explode for exactly the same reason — surface area.

Detailed Explanation

Definition

A catalyst is a substance that alters the rate of a reaction without being consumed in it. Activation energy is the minimum energy that colliding particles must possess for a reaction to occur.

Key Idea

A catalyst does not supply energy; it provides an alternative pathway with lower activation energy, so a much larger fraction of collisions become effective.

Working Principle

Raising the temperature by 10 °C typically doubles the rate, because it both increases collision frequency and, far more importantly, sharply increases the fraction of molecules with energy above the activation energy.

Applications

- Finely divided iron catalyses the Haber process for ammonia.
- Enzymes are biological catalysts that work under mild conditions.
- Refrigeration slows food spoilage by reducing the rate of microbial reactions.

✓ Points to Remember (Exam Focus)

✓ Increasing concentration or pressure increases the frequency of collisions and hence the rate.

✓ Increasing surface area (powdering a solid) increases the rate.

✓ A 10 °C rise in temperature roughly doubles the rate of many reactions.

✓ A catalyst lowers the activation energy and is not consumed.

✓ A catalyst speeds up the forward and reverse reactions equally and cannot shift the equilibrium position.

⚠ Common Student Mistakes

⚠ Saying a catalyst supplies energy to the reaction; it lowers the energy barrier instead.

⚠ Believing a catalyst increases the yield of a reaction; it only changes how fast equilibrium is reached.

⚠ Assuming temperature affects rate only by making particles move faster; the energy distribution matters far more.

Memory Hook

'Hotter, Stronger, Finer, Catalysed' — the four ways to speed a reaction up.

Advanced Insights

The Arrhenius equation k = Ae^(−Ea/RT) shows why temperature has such a dramatic effect: activation energy sits in an exponential. A small rise in T produces a disproportionately large rise in the fraction of successful collisions.

⏸ Quick Recap — Topics Covered So Far

- Rate of Reaction and Its Measurement — The rate of a reaction is the change in concentration of a reactant or product per unit time, and it decreases as the reaction proceeds and reactants are used up.
- Factors Affecting the Rate of Reaction — Reaction rate depends on the nature of reactants, concentration, surface area, temperature, catalyst and, for gases, pressure.

6.3 Collision Theory and Activation Energy

⏱ **Estimated time:** 14 mins **Difficulty:** 🔴 Hard ★★★ **Exam Importance:** ⭐⭐⭐⭐

Quick Summary

Reactions occur when particles collide with sufficient energy and correct orientation; the minimum energy required is the activation energy.

Real-Life Example

A mixture of petrol vapour and air sits harmlessly in a cylinder until a spark supplies the activation energy — then it burns in milliseconds.

📖 Detailed Explanation

Definition

Collision theory states that for a reaction to occur, reacting particles must collide with energy equal to or greater than the activation energy and with the proper orientation. An effective collision is one that results in a reaction.

Key Idea

Most collisions are ineffective. Only the small fraction that are both energetic enough and correctly oriented lead to reaction, which is why reactions are so much slower than collision frequency alone would suggest.

Working Principle

The activated complex or transition state is the unstable arrangement at the top of the energy barrier. In an exothermic reaction the products lie below the reactants in energy; in an endothermic reaction they lie above.

Applications

- Energy profile diagrams show reactants, activation energy, transition state and products.
- A catalyst lowers the peak of the barrier without changing the energy of reactants or products.
- The overall energy change ΔH is unaffected by the presence of a catalyst.

✓ Points to Remember (Exam Focus)

✓ Two conditions for an effective collision: sufficient energy AND correct orientation.

✓ Activation energy is the minimum energy needed for a reaction to occur.

✓ The activated complex is the highest-energy arrangement along the reaction path.

✓ Exothermic: products lower in energy than reactants, ΔH negative.

✓ A catalyst lowers activation energy but does not change ΔH.

⚠ Common Student Mistakes

⚠ Assuming every collision produces a reaction; the vast majority do not.

⚠ Forgetting the orientation requirement and citing energy alone.

⚠ Thinking a catalyst changes the enthalpy change of a reaction; it does not.

Memory Hook

'Enough energy AND the right angle' — both conditions, or nothing happens.

Advanced Insights

Orientation matters so much that some enzyme-catalysed reactions run millions of times faster than the uncatalysed version simply because the enzyme holds the substrates in exactly the right alignment.

Quick Recap — Topics Covered So Far

- Collision Theory and Activation Energy — Reactions occur when particles collide with sufficient energy and correct orientation; the minimum energy required is the activation energy.

Chapter 6 — End of Chapter Revision

All Memory Hooks

- Rate of Reaction and Its Measurement: 'Rate = how fast the concentration changes' — highest at the start, falling ever after.
- Factors Affecting the Rate of Reaction: 'Hotter, Stronger, Finer, Catalysed' — the four ways to speed a reaction up.
- Collision Theory and Activation Energy: 'Enough energy AND the right angle' — both conditions, or nothing happens.

⚠ All Common Mistakes

⚠ Reporting a negative rate for a reactant; the negative sign is built into the definition.

⚠ Ignoring the stoichiometric coefficients when relating rates of different species.

⚠ Confusing average rate with instantaneous rate.

⚠ Saying a catalyst supplies energy to the reaction; it lowers the energy barrier instead.

⚠ Believing a catalyst increases the yield of a reaction; it only changes how fast equilibrium is reached.

⚠ Assuming temperature affects rate only by making particles move faster; the energy distribution matters far more.

⚠ Assuming every collision produces a reaction; the vast majority do not.

⚠ Forgetting the orientation requirement and citing energy alone.

⚠ Thinking a catalyst changes the enthalpy change of a reaction; it does not.

Formula Sheet

Formula	Meaning / Use
Average Rate of Reaction: Rate = Δ[concentration] / Δt	To find how fast a reactant is consumed or a product formed over a measured interval.

Quick Revision Section

- Rate of Reaction and Its Measurement: The rate of a reaction is the change in concentration of a reactant or product per unit time, and it decreases as the reaction proceeds and reactants are used up.
- Factors Affecting the Rate of Reaction: Reaction rate depends on the nature of reactants, concentration, surface area, temperature, catalyst and, for gases, pressure.

- Collision Theory and Activation Energy: Reactions occur when particles collide with sufficient energy and correct orientation; the minimum energy required is the activation energy.

Chapter 7: Electrochemistry

Prerequisites

Before starting this chapter, make sure you are comfortable with the following ideas:

- Understanding of oxidation and reduction as electron transfer.
- Knowledge of ionic compounds and how they conduct.
- Familiarity with the reactivity series.
- Basic idea of electric current and a circuit.

7.1 Electrolytes and Electrolysis

⏱ **Estimated time:** 16 mins **Difficulty:** 🟡 Medium ★★☆ **Exam Importance:** ⭐⭐⭐⭐⭐

🔑 **Quick Summary**

Electrolytes conduct electricity in molten or aqueous state because they contain free ions, and electrolysis is the decomposition of an electrolyte by passing current through it.

🌍 **Real-Life Example**

Electroplating a steel spoon with silver, and extracting aluminium from bauxite, are the same process at two very different scales.

📖 Detailed Explanation

Definition

An electrolyte is a substance that conducts electricity in molten or aqueous state and is decomposed by it. Electrolysis is the chemical decomposition of an electrolyte by the passage of an electric current.

Key Idea

Cations move to the cathode and are reduced by gaining electrons; anions move to the anode and are oxidised by losing electrons. Reduction always happens at the cathode, whatever the type of cell.

Working Principle

Where more than one ion could be discharged, the one lower in the electrochemical series is discharged preferentially. In aqueous solutions, water may be discharged instead of a very reactive metal ion.

Applications

- Electroplating: the article is the cathode and the plating metal is the anode.
- Electrolytic refining of copper: impure copper anode, pure copper cathode.
- Extraction of sodium, aluminium, magnesium and calcium from their molten ores.

✓ Points to Remember (Exam Focus)

✓ Strong electrolytes ionise almost completely (NaCl, HCl, NaOH); weak ones only partly (CH_3COOH, NH_4OH).

✓ Cathode is negative in electrolysis and attracts cations; anode is positive and attracts anions.

✓ Reduction occurs at the cathode; oxidation at the anode — in every cell.

✓ Non-electrolytes such as sugar and urea do not conduct even in solution.

✓ Electrolysis of brine gives NaOH, Cl_2 and H_2 (the chlor-alkali process).

⚠ **Common Student Mistakes**

⚠ Saying the cathode is positive; in electrolysis it is the negative electrode.

⚠ Confusing strong electrolyte (fully ionised) with concentrated solution (large amount dissolved).

⚠ Forgetting that reduction is always at the cathode, regardless of electrode sign.

Memory Hook

'CAT-ions go to the CAThode; AN-ions go to the ANode' — and 'Red Cat, An Ox': Reduction at Cathode, Oxidation at Anode.

Advanced Insights

In electrolysis of aqueous sodium chloride, hydrogen is discharged rather than sodium because water is more easily reduced. Product prediction therefore depends on the electrochemical series, not merely on which ions are present.

7.2 Electrochemical Cells and Cell Reactions

⏱ **Estimated time:** 15 mins **Difficulty:** 🟡 Medium ★★☆ **Exam Importance:** ⭐⭐⭐⭐⭐

Quick Summary

An electrochemical cell converts chemical energy into electrical energy through a spontaneous redox reaction, with oxidation at the anode and reduction at the cathode.

Real-Life Example

Every torch cell, mobile phone battery and car battery is a packaged redox reaction whose electrons have been made to travel through a wire.

📖 Detailed Explanation

Definition

A galvanic or voltaic cell converts chemical energy into electrical energy using a spontaneous redox reaction. An electrolytic cell does the reverse, using electrical energy to drive a non-spontaneous reaction.

Key Idea

The sign of the electrodes reverses between the two cell types, but the chemistry does not: oxidation is always at the anode and reduction always at the cathode. Memorising the chemistry rather than the signs avoids all confusion.

Working Principle

In a Daniell cell, zinc is oxidised at the anode and copper ions are reduced at the cathode. A salt bridge completes the circuit and maintains electrical neutrality in both half-cells.

Applications

- Dry cell (Leclanché): zinc anode, graphite cathode with MnO_2 — used in torches and clocks.
- Lead storage battery: rechargeable, used in cars.
- Fuel cells combine hydrogen and oxygen to give electricity with water as the only product.

✓ Points to Remember (Exam Focus)

- ✓ Galvanic cell: chemical → electrical energy; anode negative, cathode positive.
- ✓ Electrolytic cell: electrical → chemical energy; anode positive, cathode negative.
- ✓ Oxidation at anode and reduction at cathode in BOTH types of cell.
- ✓ The salt bridge maintains electrical neutrality and completes the circuit.
- ✓ Electrons flow from anode to cathode through the external wire.

⚠ Common Student Mistakes

- ⚠ Assuming the anode is always positive; it is negative in a galvanic cell.
- ⚠ Saying the salt bridge carries electrons; it carries ions.
- ⚠ Reversing the direction of electron flow in the external circuit.

Memory Hook

'An Ox and a Red Cat' — ANode OXidation, REDuction CAThode, in every cell ever built.

Advanced Insights

A fuel cell is not a battery: reactants are supplied continuously rather than stored, so it never runs down. Hydrogen fuel cells achieve efficiencies well above heat engines because they are not limited by the second law's temperature bound.

Quick Recap — Topics Covered So Far

- Electrolytes and Electrolysis — Electrolytes conduct electricity in molten or aqueous state because they contain free ions, and electrolysis is the decomposition of an electrolyte by passing current through it.
- Electrochemical Cells and Cell Reactions — An electrochemical cell converts chemical energy into electrical energy through a spontaneous redox reaction, with oxidation at the anode and reduction at the cathode.

7.3 Faraday's Laws and Applications

⏱ **Estimated time:** 14 mins **Difficulty:** 🔴 Hard ★★★ **Exam Importance:** ⭐⭐⭐⭐

Quick Summary

Faraday's laws relate the mass deposited during electrolysis to the quantity of charge passed and to the equivalent mass of the substance.

Real-Life Example

The thickness of gold on a plated ornament is controlled precisely by choosing the current and the time — Faraday's first law used commercially.

📖 Detailed Explanation

Definition

Faraday's first law states that the mass of a substance deposited at an electrode is directly proportional to the quantity of electricity passed. The second law states that when the same quantity of electricity passes through different electrolytes, the masses deposited are proportional to their equivalent masses.

Key Idea

One faraday, 96500 coulombs, is the charge on one mole of electrons. It deposits exactly one gram-equivalent of any substance, which is what makes the second law work across different electrolytes.

Working Principle

Charge $Q = I \times t$, so mass deposited $m = ZIt$, where Z is the electrochemical equivalent. Doubling either the current or the time doubles the mass deposited.

Applications

- Calculating the time needed to deposit a required thickness in electroplating.
- Determining the purity of a metal by electrolytic refining.
- Estimating the charge on an electron from electrolysis measurements.

✓ Points to Remember (Exam Focus)

✓ 1 faraday = 96500 C = charge on one mole of electrons.

✓ $Q = I \times t$, where I is current in amperes and t is time in seconds.

✓ m = Z I t, where Z is the electrochemical equivalent in g/C.

✓ Equivalent mass = atomic mass / valency.

✓ Equal charge deposits equal numbers of gram-equivalents of different substances.

⚠ Common Student Mistakes

⚠ Using time in minutes instead of seconds when calculating charge.

⚠ Using atomic mass instead of equivalent mass in the second law.

⚠ Forgetting to divide by valency when finding the equivalent mass.

Memory Hook

'Charge = Current × Time; Mass follows Charge' — and one faraday, one gram-equivalent.

Advanced Insights

Faraday's laws were among the first quantitative evidence that electricity itself is particulate. The fixed charge per mole implied a fundamental unit of charge decades before the electron was actually discovered.

Formula Section

Faraday's First Law

Difficulty	🟡 Medium ★★☆
Formula	m = Z I t
Meaning of Variables	m = mass deposited in g, Z = electrochemical equivalent in g/C, I = current in A, t = time in s.
When to Use	To find the mass deposited or the time needed in any electroplating or refining problem.
Common Mistakes	Using time in minutes rather than seconds.
Shortcut / Memory Trick	Z = equivalent mass / 96500.

Quantity of Electricity

Difficulty	🟢 Easy ★☆☆
Formula	Q = I × t
Meaning of Variables	Q = charge in coulombs, I = current in amperes, t = time in seconds.

When to Use	As the first step in every Faraday's law calculation.
Common Mistakes	Mixing units of time.
Shortcut / Memory Trick	Divide Q by 96500 to get the number of faradays passed.

II Quick Recap — Topics Covered So Far

- Faraday's Laws and Applications — Faraday's laws relate the mass deposited during electrolysis to the quantity of charge passed and to the equivalent mass of the substance.

Chapter 7 — End of Chapter Revision

All Memory Hooks

- Electrolytes and Electrolysis: 'CAT-ions go to the CAThode; AN-ions go to the ANode' — and 'Red Cat, An Ox': Reduction at Cathode, Oxidation at Anode.
- Electrochemical Cells and Cell Reactions: 'An Ox and a Red Cat' — ANode OXidation, REDuction CAThode, in every cell ever built.
- Faraday's Laws and Applications: 'Charge = Current × Time; Mass follows Charge' — and one faraday, one gram-equivalent.

⚠ All Common Mistakes

⚠ Saying the cathode is positive; in electrolysis it is the negative electrode.

⚠ Confusing strong electrolyte (fully ionised) with concentrated solution (large amount dissolved).

⚠ Forgetting that reduction is always at the cathode, regardless of electrode sign.

⚠ Assuming the anode is always positive; it is negative in a galvanic cell.

⚠ Saying the salt bridge carries electrons; it carries ions.

⚠ Reversing the direction of electron flow in the external circuit.

⚠ Using time in minutes instead of seconds when calculating charge.

⚠ Using atomic mass instead of equivalent mass in the second law.

⚠ Forgetting to divide by valency when finding the equivalent mass.

Formula Sheet

Formula	Meaning / Use
Faraday's First Law: $m = Z I t$	To find the mass deposited or the time needed in any electroplating or refining problem.
Quantity of Electricity: $Q = I \times t$	As the first step in every Faraday's law calculation.

Quick Revision Section

- Electrolytes and Electrolysis: Electrolytes conduct electricity in molten or aqueous state because they contain free ions, and electrolysis is the decomposition of an electrolyte by passing current through it.

- Electrochemical Cells and Cell Reactions: An electrochemical cell converts chemical energy into electrical energy through a spontaneous redox reaction, with oxidation at the anode and reduction at the cathode.
- Faraday's Laws and Applications: Faraday's laws relate the mass deposited during electrolysis to the quantity of charge passed and to the equivalent mass of the substance.

Chapter 8: General Organic Chemistry — Nomenclature and Isomerism

Prerequisites

Before starting this chapter, make sure you are comfortable with the following ideas:

- Tetravalency of carbon and covalent bonding.
- Knowledge of homologous series and hydrocarbons.
- Familiarity with functional groups such as −OH and −COOH.
- Ability to draw simple structural formulae.

8.1 IUPAC Nomenclature of Organic Compounds

⏱ **Estimated time:** 17 mins **Difficulty:** 🟡 Medium ★★☆ **Exam Importance:** ⭐⭐⭐⭐⭐

🔎 Quick Summary

An IUPAC name is built from a root indicating the longest carbon chain, a suffix for the functional group and prefixes for the substituents, each with a locant number.

🌍 Real-Life Example

The common name 'wood spirit' tells you nothing, but 'methanol' tells you at once that it has one carbon and an −OH group.

📖 Detailed Explanation

Definition

The IUPAC name of an organic compound consists of a word root (number of carbon atoms in the longest chain), a primary suffix (type of bond), a secondary suffix (functional group) and prefixes (substituents).

Key Idea

Numbering must give the lowest possible locant to the principal functional group first, then to double or triple bonds, and only then to substituents. Getting this priority order right resolves nearly every naming dispute.

Working Principle

Select the longest continuous chain containing the functional group, number it from the end nearer that group, name substituents alphabetically with their locants, and assemble prefix–root–suffix.

Applications

- CH_3-CH_2-OH is ethanol: eth (2 carbons) + an (single bonds) + ol (alcohol).
- CH_3-CHO is ethanal; CH_3-COOH is ethanoic acid; $CH_3-CO-CH_3$ is propanone.
- Branched chains: $(CH_3)_2CH-CH_3$ is 2-methylpropane.

✓ Points to Remember (Exam Focus)

- ✓ Word roots: meth (1), eth (2), prop (3), but (4), pent (5), hex (6), hept (7), oct (8).
- ✓ Primary suffix: -ane (single), -ene (double), -yne (triple).
- ✓ Secondary suffix: -ol (alcohol), -al (aldehyde), -one (ketone), -oic acid (carboxylic acid).
- ✓ The final 'e' of the primary suffix is dropped before a suffix beginning with a vowel.
- ✓ Number the chain to give the functional group the lowest possible locant.

⚠ Common Student Mistakes

- ⚠ Numbering from the wrong end and giving the functional group a higher locant.
- ⚠ Choosing the longest chain without checking that it contains the functional group.
- ⚠ Retaining the final 'e' — writing 'ethaneol' instead of 'ethanol'.

Memory Hook

'Prefix–Root–Suffix' and 'functional group gets the lowest number, always'.

Advanced Insights

IUPAC nomenclature is a generative grammar: a finite set of rules names an unbounded set of compounds unambiguously. This is why chemical databases can be searched by name across millions of structures.

8.2 Structural Isomerism

⏱ **Estimated time:** 16 mins **Difficulty:** 🟡 Medium ★★☆ **Exam Importance:** ⭐⭐⭐⭐⭐

Quick Summary

Isomers have the same molecular formula but different structures, and structural isomerism includes chain, position and functional isomerism.

Real-Life Example

Butane and isobutane have identical formulas but different boiling points, which is why one liquefies more easily and is preferred in cold-climate LPG.

Detailed Explanation

Definition

Isomers are compounds with the same molecular formula but different structural formulae, and therefore different properties. Chain isomers differ in the carbon skeleton, position isomers in the position of a group or bond, and functional isomers in the functional group itself.

Key Idea

Isomerism is a direct consequence of carbon's tetravalency and catenation. The more carbon atoms a molecule has, the more ways they can be arranged, which is why isomer counts explode with chain length.

Working Principle

To find all isomers systematically, first draw the longest straight chain, then progressively shorten it and place the removed carbons as branches, checking for duplicates by rotation.

Applications

- C_4H_{10} has two chain isomers: butane and 2-methylpropane.
- C_3H_8O shows position isomerism: propan-1-ol and propan-2-ol.
- C_2H_6O shows functional isomerism: ethanol (alcohol) and dimethyl ether.

✓ Points to Remember (Exam Focus)

✓ Isomers have the same molecular formula but different structures and different properties.

✓ Chain isomerism starts from butane (C_4H_{10}); the first three alkanes have no isomers.

✓ Position isomerism: same skeleton, different position of the functional group.

✓ Functional isomerism: same formula, entirely different functional group.

✓ The number of possible isomers rises very steeply with the number of carbon atoms.

⚠ Common Student Mistakes

⚠ Counting the same structure twice after simply rotating or redrawing it.

⚠ Claiming methane, ethane or propane have isomers; they do not.

⚠ Confusing isomers (same formula) with homologues (differing by CH_2).

Memory Hook

'Same formula, different arrangement' — Chain, Position, Functional: three ways to rearrange.

Advanced Insights

Stereoisomers have the same connectivity but differ in spatial arrangement, and can differ dramatically in biological effect. The tragedy of thalidomide, where one mirror-image form was a sedative and the other caused birth defects, made stereochemistry a regulatory requirement.

Quick Recap — Topics Covered So Far

- IUPAC Nomenclature of Organic Compounds — An IUPAC name is built from a root indicating the longest carbon chain, a suffix for the functional group and prefixes for the substituents, each with a locant number.

- Structural Isomerism — Isomers have the same molecular formula but different structures, and structural isomerism includes chain, position and functional isomerism.

8.3 Functional Groups and Homologous Series

⏱ **Estimated time:** 14 mins **Difficulty:** 🟡 Medium ★★☆ **Exam Importance:** ⭐⭐⭐⭐⭐

🔑 Quick Summary

A functional group is the atom or group that determines the chemical properties of an organic compound, and compounds sharing one functional group form a homologous series.

🌍 Real-Life Example

Every alcohol from methanol to the alcohol in hand sanitiser reacts with sodium to release hydrogen, because they all carry the same −OH group.

📖 Detailed Explanation

Definition

A functional group is an atom or group of atoms that gives characteristic chemical properties to an organic compound. A homologous series is a group of compounds with the same functional group and general formula, successive members differing by $-CH_2-$.

Key Idea

Chemistry is decided by the functional group and physical properties by the chain length. This division is what makes organic chemistry learnable — you learn a group once and apply it to a whole family.

Working Principle

As molecular mass increases along a series, boiling point and viscosity increase steadily while solubility in water decreases, because the non-polar hydrocarbon part grows relative to the polar group.

Applications

- Alcohols (−OH): react with sodium to give hydrogen; oxidised to carboxylic acids.
- Carboxylic acids (−COOH): turn blue litmus red; give CO_2 with sodium carbonate.
- Aldehydes (−CHO) are easily oxidised; ketones (>C=O) are not.

✓ Points to Remember (Exam Focus)

- ✓ Members of a homologous series differ by CH_2 and by 14 u in molecular mass.
- ✓ All members share the same functional group and hence similar chemical properties.
- ✓ Physical properties change gradually with increasing molecular mass.
- ✓ Boiling point increases and water solubility decreases as the chain lengthens.
- ✓ Functional group determines chemistry; chain length determines physical behaviour.

⚠ Common Student Mistakes

⚠ Saying members of a homologous series have identical properties; chemical properties are similar, physical ones differ.

⚠ Confusing aldehyde (−CHO, at the chain end) with ketone (>C=O, within the chain).

⚠ Assuming all alcohols are soluble in water; solubility falls sharply beyond four carbons.

Memory Hook

'Group decides the chemistry, chain decides the physics.'

Advanced Insights

The −COOH group is acidic while −OH in alcohols is essentially neutral, because the carboxylate ion left after losing H^+ is stabilised by resonance. Resonance stabilisation of the product, not the strength of the O−H bond, decides acidity.

Quick Recap — Topics Covered So Far

- Functional Groups and Homologous Series — A functional group is the atom or group that determines the chemical properties of an organic compound, and compounds sharing one functional group form a homologous series.

Chapter 8 — End of Chapter Revision

All Memory Hooks

- IUPAC Nomenclature of Organic Compounds: 'Prefix–Root–Suffix' and 'functional group gets the lowest number, always'.
- Structural Isomerism: 'Same formula, different arrangement' — Chain, Position, Functional: three ways to rearrange.
- Functional Groups and Homologous Series: 'Group decides the chemistry, chain decides the physics.'

⚠ All Common Mistakes

⚠ Numbering from the wrong end and giving the functional group a higher locant.

⚠ Choosing the longest chain without checking that it contains the functional group.

⚠ Retaining the final 'e' — writing 'ethaneol' instead of 'ethanol'.

⚠ Counting the same structure twice after simply rotating or redrawing it.

⚠ Claiming methane, ethane or propane have isomers; they do not.

⚠ Confusing isomers (same formula) with homologues (differing by CH_2).

⚠ Saying members of a homologous series have identical properties; chemical properties are similar, physical ones differ.

⚠ Confusing aldehyde (–CHO, at the chain end) with ketone (>C=O, within the chain).

⚠ Assuming all alcohols are soluble in water; solubility falls sharply beyond four carbons.

Quick Revision Section

- IUPAC Nomenclature of Organic Compounds: An IUPAC name is built from a root indicating the longest carbon chain, a suffix for the functional group and prefixes for the substituents, each with a locant number.
- Structural Isomerism: Isomers have the same molecular formula but different structures, and structural isomerism includes chain, position and functional isomerism.
- Functional Groups and Homologous Series: A functional group is the atom or group that determines the chemical properties of an organic compound, and compounds sharing one functional group form a homologous series.

Chapter 9: Organic Compounds

Prerequisites

Before starting this chapter, make sure you are comfortable with the following ideas:

- IUPAC nomenclature and functional groups.
- Knowledge of addition and substitution reactions.
- Understanding of oxidation of alcohols.
- Familiarity with soaps, detergents and esters.

9.1 Alkanes, Alkenes and Alkynes

⏱ **Estimated time:** 16 mins **Difficulty:** 🟡 Medium ★★☆ **Exam Importance:** ⭐⭐⭐⭐⭐

🔑 Quick Summary

Alkanes are saturated and undergo substitution, while alkenes and alkynes are unsaturated and undergo addition, which makes them far more reactive.

🌍 Real-Life Example

Ethyne burning in an oxy-acetylene torch reaches over 3000 °C, hot enough to cut steel — energy released from breaking a triple bond.

📖 Detailed Explanation

Definition

Alkanes (C_nH_{2n+2}) contain only C–C single bonds. Alkenes (C_nH_{2n}) contain at least one C=C double bond. Alkynes (C_nH_{2n-2}) contain at least one C≡C triple bond.

Key Idea

The multiple bond is a site of high electron density and is therefore attacked readily, which is why unsaturated hydrocarbons are far more reactive than alkanes despite the multiple bond being stronger overall.

Working Principle

Alkanes react by free-radical substitution in sunlight, giving a mixture of products. Alkenes and alkynes react by addition across the multiple bond, giving a single major product.

Applications

- Bromine water test: unsaturated compounds decolourise reddish-brown bromine water; alkanes do not.
- Ethene is used to ripen fruit artificially and to make polythene.
- Ethyne is used in oxy-acetylene welding and cutting.

✓ Points to Remember (Exam Focus)

✓ Alkanes: substitution reactions, require sunlight, give a mixture of products.
✓ Alkenes and alkynes: addition reactions with H_2, Br_2, HCl and H_2O.
✓ Bromine water test distinguishes saturated from unsaturated hydrocarbons.
✓ Unsaturated hydrocarbons burn with a sooty yellow flame; saturated ones with a blue flame.
✓ Alkanes are relatively unreactive and were once called paraffins, meaning 'little affinity'.

⚠ Common Student Mistakes

- ⚠ Expecting alkanes to decolourise bromine water; only unsaturated compounds do.
- ⚠ Writing the alkyne general formula as C_nH_{2n-1}.
- ⚠ Assuming substitution gives a single clean product; it gives a mixture.

Memory Hook

'Saturated Substitutes, Unsaturated Adds' — and 'bromine water fades only for the unsaturated'.

Advanced Insights

Addition to an unsymmetrical alkene follows Markovnikov's rule: the hydrogen adds to the carbon already carrying more hydrogens. The rule reflects the stability of the intermediate carbocation, not any preference of the reagent.

9.2 Alcohols, Aldehydes and Ketones

⏱ **Estimated time:** 16 mins **Difficulty:** 🟡 Medium ★★☆ **Exam Importance:** ⭐⭐⭐⭐⭐

Quick Summary

Alcohols carry the −OH group and are oxidised first to aldehydes or ketones and then, for primary alcohols, to carboxylic acids.

Real-Life Example

The unpleasant symptoms of a hangover come largely from ethanal, the aldehyde produced when the liver oxidises ethanol.

📖 Detailed Explanation

Definition

Alcohols contain the hydroxyl group −OH. Aldehydes contain −CHO at the end of a chain; ketones contain >C=O within the chain. Denatured alcohol is ethanol made unfit for drinking by adding methanol and a dye.

Key Idea

Primary alcohols oxidise to aldehydes and then to carboxylic acids; secondary alcohols oxidise to ketones and stop there. The position of the −OH group therefore determines the whole oxidation pathway.

Working Principle

Alcohols react with sodium to liberate hydrogen, which distinguishes them from hydrocarbons. Concentrated sulphuric acid at 443 K dehydrates ethanol to ethene.

Applications

- Ethanol: alcoholic drinks, solvent, antiseptic, tincture of iodine, fuel additive.
- Methanol is highly toxic — it causes blindness and death even in small quantities.
- Aldehydes reduce Tollens' reagent (silver mirror) and Fehling's solution; ketones do not.

✓ Points to Remember (Exam Focus)

✓ Alcohol + Na → sodium alkoxide + H_2 (test for the −OH group).

✓ Primary alcohol → aldehyde → carboxylic acid; secondary alcohol → ketone only.

✓ Oxidising agents: alkaline $KMnO_4$, acidified $K_2Cr_2O_7$.

✓ Tollens' and Fehling's tests distinguish aldehydes from ketones.

✓ Ethanol dehydrated by conc. H_2SO_4 at 443 K gives ethene.

⚠ Common Student Mistakes

⚠ Saying ketones give a positive Tollens' test; only aldehydes do.

⚠ Confusing dehydration (removes water, gives alkene) with oxidation (adds oxygen, gives acid).

⚠ Treating methanol as a harmless substitute for ethanol; it is severely toxic.

Memory Hook

'Primary goes all the way to acid, Secondary stops at ketone.'

Advanced Insights

Methanol poisoning is treated with ethanol, which competes for the same enzyme and prevents methanol being converted into formic acid. It is a rare case where an antidote works by occupying the enzyme rather than neutralising the poison.

Quick Recap — Topics Covered So Far

- Alkanes, Alkenes and Alkynes — Alkanes are saturated and undergo substitution, while alkenes and alkynes are unsaturated and undergo addition, which makes them far more reactive.

- Alcohols, Aldehydes and Ketones — Alcohols carry the −OH group and are oxidised first to aldehydes or ketones and then, for primary alcohols, to carboxylic acids.

9.3 Carboxylic Acids and Esters

⏱ **Estimated time:** 15 mins **Difficulty:** 🟡 Medium ★★☆ **Exam Importance:** ⭐⭐⭐⭐⭐

🔑 Quick Summary

Carboxylic acids carry the −COOH group and are weak acids, and they react with alcohols to give sweet-smelling esters.

🌍 Real-Life Example

The smell of pineapple, banana and jasmine in sweets and perfumes usually comes from synthetic esters made in exactly this way.

📖 Detailed Explanation

Definition

Carboxylic acids contain the carboxyl group −COOH. Esterification is the reaction of a carboxylic acid with an alcohol in the presence of concentrated sulphuric acid to give an ester and water. Saponification is the hydrolysis of an ester with alkali to give soap.

Key Idea

Carboxylic acids are weak acids because they ionise only partially, yet they are strong enough to displace CO_2 from carbonates — a reaction that distinguishes them sharply from alcohols and phenols.

Working Principle

Concentrated sulphuric acid acts as both catalyst and dehydrating agent in esterification, removing the water formed and driving the equilibrium towards the ester.

Applications

- Ethanoic acid + ethanol → ethyl ethanoate, a sweet-smelling ester.
- Esters are used in perfumes, artificial flavours and as solvents.
- Saponification of vegetable oil with NaOH gives soap and glycerol.

✓ Points to Remember (Exam Focus)

- ✓ Carboxylic acids turn blue litmus red and are weak acids.
- ✓ Acid + Na_2CO_3 or $NaHCO_3$ → salt + CO_2 + water, with brisk effervescence.
- ✓ Esterification needs concentrated H_2SO_4 as catalyst and dehydrating agent.
- ✓ Esters are sweet-smelling; alkaline hydrolysis of an ester is called saponification.
- ✓ Ethanoic acid is called glacial acetic acid because it freezes at 290 K.

⚠ **Common Student Mistakes**

- ⚠ Saying carboxylic acids are strong acids; they are weak and only partially ionised.
- ⚠ Expecting alcohols to give CO_2 with carbonate; only carboxylic acids do.
- ⚠ Confusing esterification with saponification — they run in opposite directions.

Memory Hook

'Acid + Alcohol = Ester + water (sweet); Ester + Alkali = Soap (saponification).'

Advanced Insights

Carboxylic acids are acidic because the carboxylate ion delocalises its negative charge over two oxygen atoms. Alcohols have no such stabilisation, which is why an −OH on a carbon chain is neutral but an −OH on a carbonyl is acidic.

9.4 Polymers and Biomolecules

⏱ **Estimated time:** 15 mins **Difficulty:** 🟡 Medium ★★☆ **Exam Importance:** ⭐⭐⭐⭐

Quick Summary

Polymers are giant molecules built from repeating monomer units, and the major biomolecules — carbohydrates, proteins, fats and nucleic acids — are natural polymers or their assemblies.

Real-Life Example

Cotton, wool, silk, starch and DNA are all natural polymers, and polythene, nylon and PVC are the synthetic answers to them.

📖 Detailed Explanation

Definition

A polymer is a large molecule formed by joining many small repeating units called monomers. Addition polymers form without loss of any small molecule; condensation polymers form with the elimination of a small molecule such as water.

Key Idea

Biomolecules are simply polymers with biological jobs: carbohydrates are polymers of sugars, proteins of amino acids, and nucleic acids of nucleotides. The same chemistry underlies a plastic bag and a strand of DNA.

Working Principle

Proteins are formed by peptide bonds between amino acids; their function depends on the precise three-dimensional folding, which is destroyed by heat or acid — a process called denaturation.

Applications

- Addition polymers: polythene, PVC, polystyrene, Teflon.
- Condensation polymers: nylon, terylene, bakelite.
- Carbohydrates: glucose (monosaccharide), sucrose (disaccharide), starch and cellulose (polysaccharides).

✓ Points to Remember (Exam Focus)

✓ Addition polymerisation adds monomers with no by-product; condensation eliminates a small molecule.

✓ Proteins are polymers of amino acids joined by peptide bonds.

✓ Starch and cellulose are both glucose polymers, but humans can digest only starch.

✓ Vitamins are classified as fat-soluble (A, D, E, K) and water-soluble (B group, C).

✓ Denaturation destroys a protein's shape and hence its function, without breaking peptide bonds.

⚠ Common Student Mistakes

⚠ Saying cellulose and starch are different sugars; both are glucose polymers linked differently.

⚠ Confusing addition polymerisation (no by-product) with condensation (water eliminated).

⚠ Thinking denaturation breaks the peptide chain; it only unfolds the structure.

Memory Hook

'Monomers make Polymers' — addition just joins, condensation joins and drops water.

Advanced Insights

Humans cannot digest cellulose because the enzyme cellulase is absent, yet ruminants can through the bacteria in their gut. The difference between food and fibre comes down to a single change in the geometry of the link between two identical glucose units.

Quick Recap — Topics Covered So Far

- Carboxylic Acids and Esters — Carboxylic acids carry the −COOH group and are weak acids, and they react with alcohols to give sweet-smelling esters.
- Polymers and Biomolecules — Polymers are giant molecules built from repeating monomer units, and the major biomolecules — carbohydrates, proteins, fats and nucleic acids — are natural polymers or their assemblies.

Chapter 9 — End of Chapter Revision

All Memory Hooks

- Alkanes, Alkenes and Alkynes: 'Saturated Substitutes, Unsaturated Adds' — and 'bromine water fades only for the unsaturated'.
- Alcohols, Aldehydes and Ketones: 'Primary goes all the way to acid, Secondary stops at ketone.'
- Carboxylic Acids and Esters: 'Acid + Alcohol = Ester + water (sweet); Ester + Alkali = Soap (saponification).'
- Polymers and Biomolecules: 'Monomers make Polymers' — addition just joins, condensation joins and drops water.

⚠ All Common Mistakes

⚠ Expecting alkanes to decolourise bromine water; only unsaturated compounds do.

⚠ Writing the alkyne general formula as C_nH_{2n-1}.

⚠ Assuming substitution gives a single clean product; it gives a mixture.

⚠ Saying ketones give a positive Tollens' test; only aldehydes do.

⚠ Confusing dehydration (removes water, gives alkene) with oxidation (adds oxygen, gives acid).

⚠ Treating methanol as a harmless substitute for ethanol; it is severely toxic.

⚠ Saying carboxylic acids are strong acids; they are weak and only partially ionised.

⚠ Expecting alcohols to give CO_2 with carbonate; only carboxylic acids do.

⚠ Confusing esterification with saponification — they run in opposite directions.

⚠ Saying cellulose and starch are different sugars; both are glucose polymers linked differently.

⚠ Confusing addition polymerisation (no by-product) with condensation (water eliminated).

⚠ Thinking denaturation breaks the peptide chain; it only unfolds the structure.

Quick Revision Section

- Alkanes, Alkenes and Alkynes: Alkanes are saturated and undergo substitution, while alkenes and alkynes are unsaturated and undergo addition, which makes them far more reactive.
- Alcohols, Aldehydes and Ketones: Alcohols carry the −OH group and are oxidised first to aldehydes or ketones and then, for primary alcohols, to carboxylic acids.
- Carboxylic Acids and Esters: Carboxylic acids carry the −COOH group and are weak acids, and they react with alcohols to give sweet-smelling esters.
- Polymers and Biomolecules: Polymers are giant molecules built from repeating monomer units, and the major biomolecules — carbohydrates, proteins, fats and nucleic acids — are natural polymers or their assemblies.